Q1.

Over a period of time it has been shown that the mean number of vehicles passing a service station on a motorway is 50 per minute.

After a new motorway junction was built nearby, Xander observed that 30 vehicles passed the service station in one minute.

(a) Xander claims that the construction of the new motorway junction has reduced the mean number of vehicles passing the service station per minute.

Investigate Xander's claim, using a suitable test at the 1% level of significance.

(6)

(b) For your test carried out in part **(a)** state, in context, the meaning of a Type 1 error.

(1)

(c) Explain why the model used in part **(a)** might be invalid.

(1)

(Total 8 marks)

Q2.

In a small town, the number of properties sold during a week in spring by a local estate agent, Keith, can be regarded as occurring independently and with constant mean μ. Data from several years have shown the value of μ to be 3.5.

A new housing development was built on the outskirts of the town and the properties on this development were offered for sale by the builder of the development, not by the local estate agents.

During the first **four** weeks in spring, when properties on the new development were offered for sale by the builder, Keith sold a total of 8 properties.

Keith claims that the sale of new properties by the builder reduced his mean number of properties sold during a week in spring.

(a) Investigate Keith's claim, using the 5% level of significance.

(6)

(b) For your test carried out in part **(a)** state, in context, the meaning of a Type II error.

(1)

(c) State **one** advantage and **one** disadvantage of using a 1% level of significance rather than a 5% level of significance in a hypothesis test.

(2)

(Total 9 marks)

Q3.

The daily number of customers visiting a small arts and crafts shop may be modelled by a Poisson distribution with a mean of 24.

The shop offers a picture framing service. The daily number of requests, Y, for this service may be assumed to have a Poisson distribution.

Prior to the shop advertising this service in the local free newspaper, the mean value of Y was 2. Following the advertisement, the shop received a total of 17 requests for the service during a period of 5 days.

(a) Using a Poisson distribution, carry out a test, at the 10% level of significance, to investigate the claim that the advertisement increased the mean daily number of requests for the shop's picture framing service.

(5)

(b) Determine the critical value of Y for your test in part (b)(i).

(3)

(c) Hence, assuming that the advertisement increased the mean value of Y to 3, determine the power of your test in part (b)(i).

(4)

(Total 12 marks)

Q4.

In a town, the total number, R, of houses sold during a week by estate agents may be modelled by a Poisson distribution with a mean of 13.

A new housing development is completed in the town. During the first week in which houses on this development are offered for sale by the developer, the estate agents sell a total of 10 houses.

(a) Using the 10% level of significance, investigate whether the offer for sale of houses by the developer has resulted in a reduction in the mean value of R.

(6)

(b) Determine, for your test in part (a), the critical region for R.

(2)

(c) Assuming that the offer for sale of houses on the new housing development has reduced the mean value of R to 6.5, determine, for a test at the 10% level of significance, the probability of a Type II error.

(4)

(Total 12 marks)

Q5.

At a hospital, the number of patients who fail to turn up for appointments on any particular day can be modelled by a Poisson distribution with mean 7.0.

The manager introduces new procedures in an attempt to improve the attendance for appointments at the hospital. On a day chosen at random following the introduction of the new procedures, it was found that only 3 patients did not turn up for their appointments.

Stating null and alternative hypotheses, test, at the 5% level of significance, the manager's claim that the new procedures have resulted in a decrease in the mean number of patients not turning up for their hospital appointments.

(Total 5 marks)

Q6.

Coloured paving stones are delivered to a builders merchant in packs. From past experience, the number of unsaleable paving stones per pack may be modelled by a Poisson distribution with a mean of 0.4.

(a) From a delivery of 20 packs, 13 paving stones are found to be unsaleable.

Investigate, at the 5% level of significance, the claim that the mean number of unsaleable paving stones per pack has increased.

(5)

(b) Subsequently, the builders merchant monitors the quality of a delivery of 100 packs, and finds a total of 52 paving stones that are unsaleable.

Using this information, re-investigate the claim made in part (a).

(6)
(Total 11 marks)

Q7.

The daily number of missed appointments at a dental surgery can be modelled by a Poisson distribution with mean 1.8.

In an attempt to reduce this figure, the surgery publicised the introduction of fines for missed appointments.

(a) During the first 5 days after its introduction, there was a total of 6 missed appointments.

Show, at the 10% level of significance, that there is insufficient evidence that the introduction of fines has succeeded in its aim.

(4)

(b) A dentist at the surgery suggested that the number of missed appointments should be monitored over a much longer period of time. Subsequently, there was a total of 297 missed appointments during 180 days.

Test, again at the 10% level of significance, the claim that the mean daily number of missed appointments is now less than 1.8.

(8)
(Total 12 marks)